ECO-SHARE: Campus Carpooling & Bike Sharing System

Submitted By:

U.Mukesh - AP23110010175
BHUMI PRASAD- AP23110010218
JEETESH KUMAR VERMA- AP23110010223
M.M.S. VIJAYA LAKSHMI - AP23110010225

BACHELOR OF TECHNOLOGY IN

Computer Science and Engineering

School of Engineering and Sciences

Under the guidance of

Mr. Himanshu Mishra

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1. Abstract

Eco-Share is a campus-based mobility platform that helps students with their commute. Students face a lot of problems when they travel to college. They have to deal with costs, traffic jams, not enough parking space and pollution. Eco-Share wants to solve these problems by offering two services: carpooling and bike-sharing.

The platform lets students share rides with friends who are going in the direction. This way students can save money. There will be fewer cars on campus. Eco-Share also has bikes that students can use for trips. This is a way to stay healthy and help the environment. Since only students with a campus account can use Eco-Share it is an trustworthy system.

Eco-Share uses the technology stack, which includes MongoDB, Express.js, React and Node.js. This makes the platform strong and able to handle a lot of users. The backend of the platform, which is powered by Node.js and Express.js handles user accounts, data and requests. MongoDB. Retrieves user and ride data. The frontend, which is built with React is easy to use and looks good. Eco-Share also has features that let users see rides, book them and plan their trips in real time.

Eco-Share is not just about making things easy and affordable for students. It also helps the environment by reducing carbon emissions and promoting shared mobility. By using technology and being eco-friendly Eco-Share is a solution, for campus transportation. It encourages students to use greener ways to commute. Eco-Share is a way for students to travel to college without hurting the environment. Eco-Share makes it easy for students to get to college without spending a lot of money. Eco-Share is a way to reduce traffic jams and pollution on campus.

2. Introduction

2.1 Motivation

Every day, commute from and around campus is a major hurdle for students. Most students have to depend on crowded public transport where time and comfort get sacrificed; the individual gets less productive. The alternative is private vehicles; which increase expense per person, contributes to congestion and creates difficulties in finding parking within campus area.

There is a massive gap in coordinating students traveling to the same destination at same time on same route through ride sharing. Even though there are a number of students traveling same route and destination at same time and hence on the same lane there is no way for them to get them on the same car pool. Hence the resources available with them were under-utilized. Short distance commute in the campus also found it difficult for students as there is no easily available and a green alternative to commute within the campus. Eco Share aims to mitigate these issues by developing an centralized platform and in the digital form which will allow the students to share their rides in an effective way. This project will thus help to decrease the cost of commute per person, will optimize the use of the available resources in use and increase greenery along with being cost-effective and convenient.

2.2Project scope

The scope of Eco Share project revolves around development of a complete campus based mobility platform including both ride and bicycle sharing mechanism. The system is designed to work in a safe environment with a defined perimeter which constitutes a campus.

The various modules of this system are;

-Real Time Ride sharing board: to facilitate the search and announcement of available car pools, allowing effective user to user ride coordination for carpooling;

-Bike rental system: to allow the student's to take a cycle on rent for small commutes within the campus;

-User authentication andRole based dashboard: A secure registration/login module which identifies the campus user and authenticates him/her in order to give role based dashboards to the driver, rider, administrator;

-Dynamic seat allocation and Trip Tracking: The system updates the number of seats in a pool and tracks an ongoing trip.

3. Problem Statement

The current ride sharing applications like Uber, Ola serve the general masses in urban areas. They are designed for broad city coverage and lack the personalization and cost-efficiency for short distance rides or the campus commute. High operational costs, dynamic surge pricing and unavailability of campus specific services are some key disadvantages.

Without a system in place, students are resorting to inefficient methods of coordinating rides, like WhatsApp groups, social media pages, and notice boards on the campus. Though flexible to some extent, this method comes with its own set of disadvantages:

Unorganized:

Information regarding the rides like seat availability, routes and timings is usually scattered all over the platform or being broadcast through messages, and it is difficult to keep track

Insecure:

These informal modes of coordination involve random users without any strict verification mechanism. This is especially a safety concern among female students.

Inefficient:

Coordination through messages and manual communication are time consuming, leading to confusion and missed rides.

Thus, a system is required which efficiently serves and covers only the students in college campus, is secure, efficient and personalized for their needs.Eco-Share is designed to fulfill this gap. The system caters specifically to a college community with unique requirements and comprises a central, automated, and secure system. Access to this system is restricted only to college-verified users.

4. Proposed System

4.1 Unique Features

The set of features provided by Eco-Share, has the following functionalities that ensures the efficiency and simplicity of the whole process and provides the best experience to the users:

Post a Ride: Users(Drivers) need to register themselves first with the necessary details like starting and ending point, trip time, number of seats available, then post the ride, other users can look up for rides from here.

Real Time Joining: The passenger user can join the available trips with a single click. The system automatically updates the seats of the vehicle. This prevents the overbooking and also provides real-time ride information.

Live Trips Board: There is a live board that shows the various rides available to each user. Users can look for rides and also find the desired routes and trip timing as required.

Bike Sharing: A set of bicycles is also available, to travel for shorter distance on the campus, which can be used with single click and the status updates. This mode of travel can also be eco-friendly.

Personalized dashboard: This feature helps in maintaining user records, the trips posted, current booked trips, history, also, it provides a control over the system for the user.

4.2 System flow

The overall flow of the Eco-Share system can be depicted as:

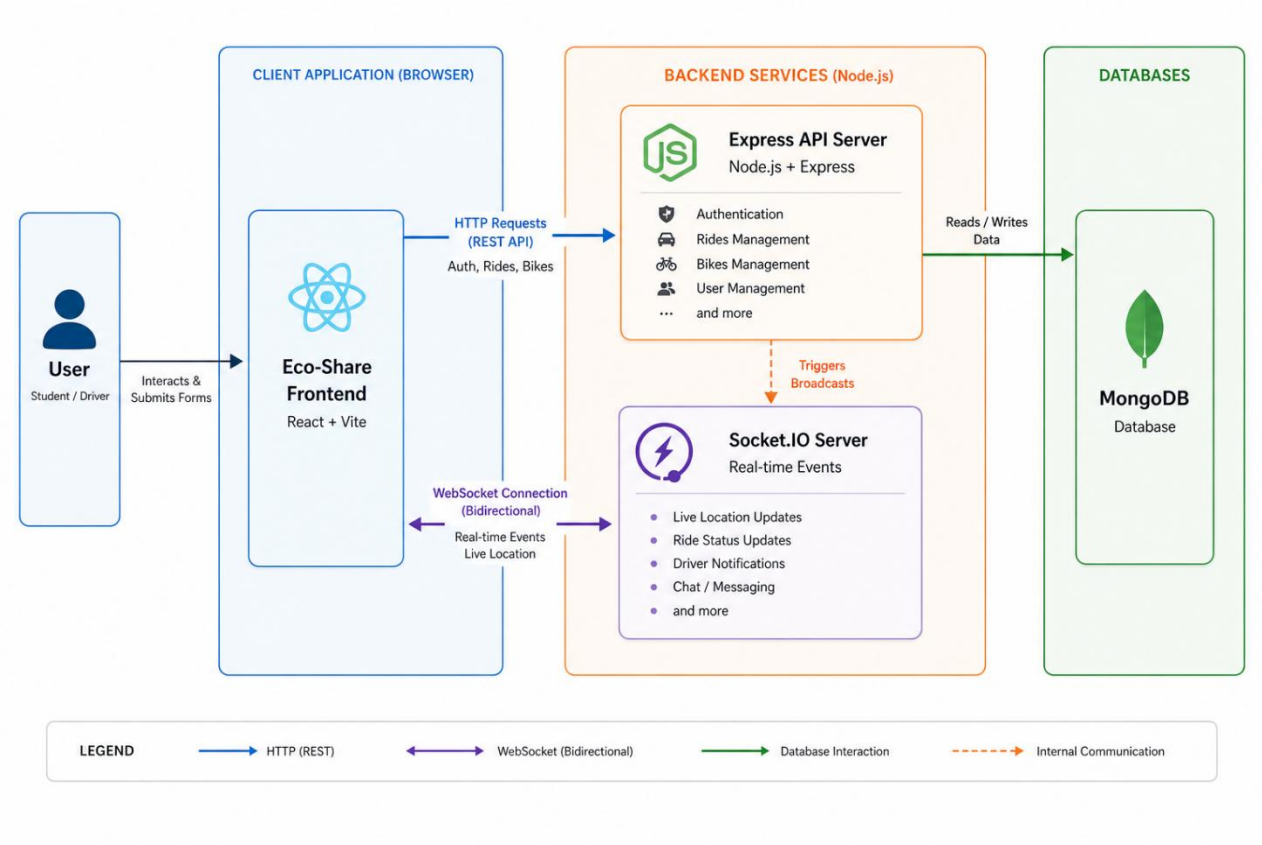
User Authentication: The first step is registration and logging in, which can only be done with a college specific email address. It makes sure that only college students have access to this system.

Discovery: In the discovery step, the users can look for various available rides from the Live Trips board and then search for specific routes accordingly.

Action: In the action step, users can book their ride as a passenger, post a new ride as a driver or book a bike to cover the shorter distance on campus.

Completion: When the trip is complete, the status is automatically updated, and a log of completed trips can be seen by the users.

5. System Architecture



5.1 Architecture Diagram (Explanation)

The Architecture of Eco-Share can be illustrated by understanding the working of different layer:

Presentation Layer (Frontend):

This is the user interface layer. The frontend is built using React.js and designed with Tailwind CSS. Users interact through this layer for performing various activities like login, posting ride, booking trips, searching rides, booking bikes and so on. The frontend will communicate with backend via Http request (API Calls).

Application Layer (Backend):

This is the core part of the application that handles the business logic, request processing, authentication and interface between the frontend and database. It is built using Node.js with the help of Express.js framework. JWT (JSON web tokens) are used here for authentication of the user.

Data Layer (Database):

The backend data is stored in the NoSQL MongoDB database. User details, rides posted, bookings, history are all stored in MongoDB. Mongoose ODM is used to define schema and access database.

Flow of Data:

User->Presentation Layer(Frontend)->Application Layer(Backend)->Data Layer(Database)->Application Layer(Backend)->Presentation Layer(Frontend)->User

5.2 Technology Stack

Eco-Share is designed and implemented using MERN Stack which helps in building efficient and scalable applications using JavaScript only, which will be efficient.

Frontend:

React 18

Vite

Tailwind CSS

Axios

Backend:

Node.js

Express.js

Database:

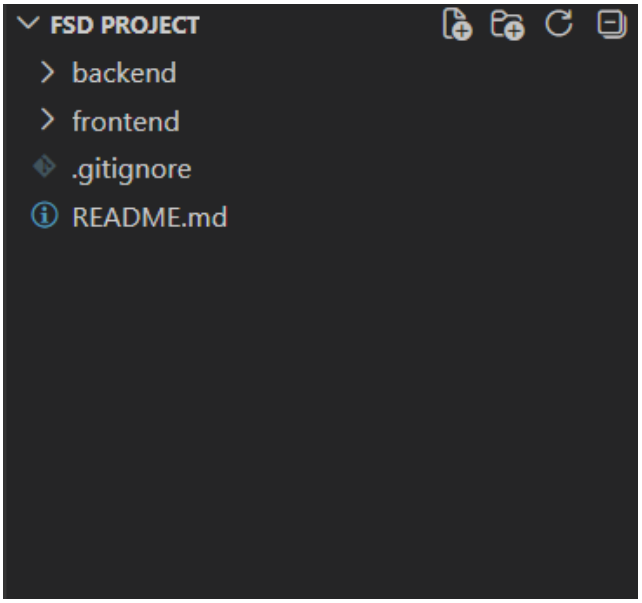
MongoDB

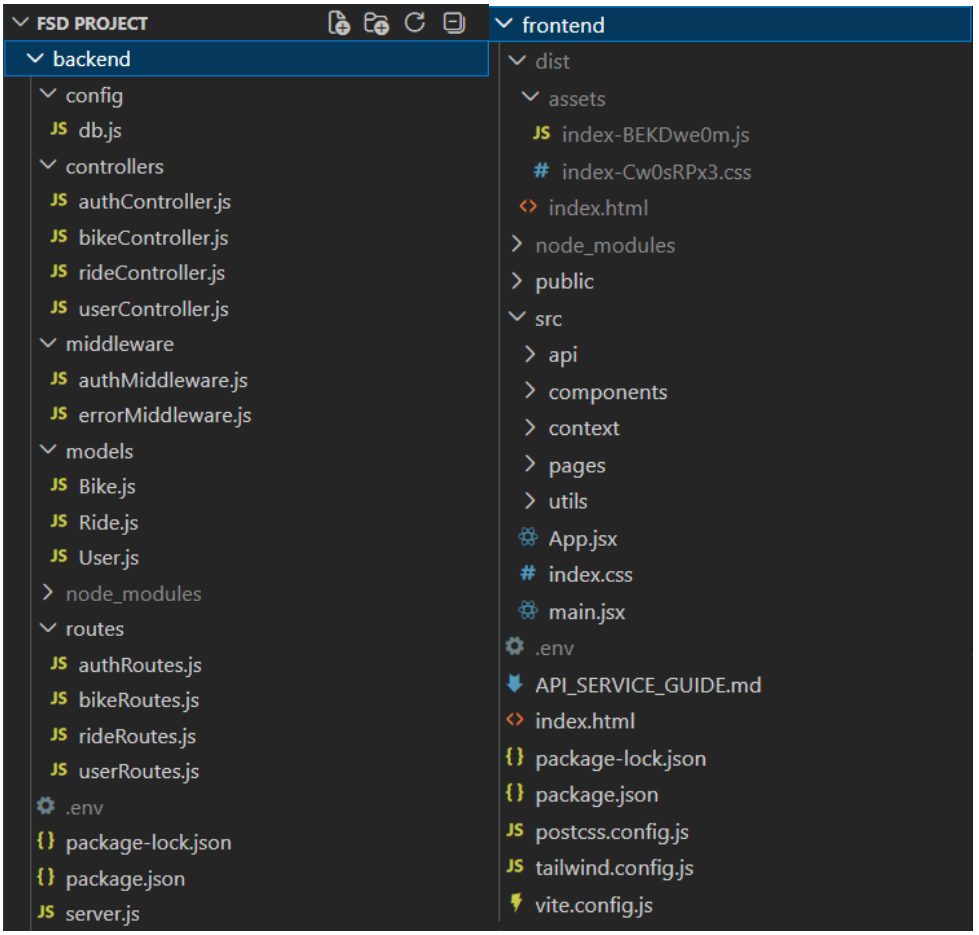
Mongoose ODM

Security:

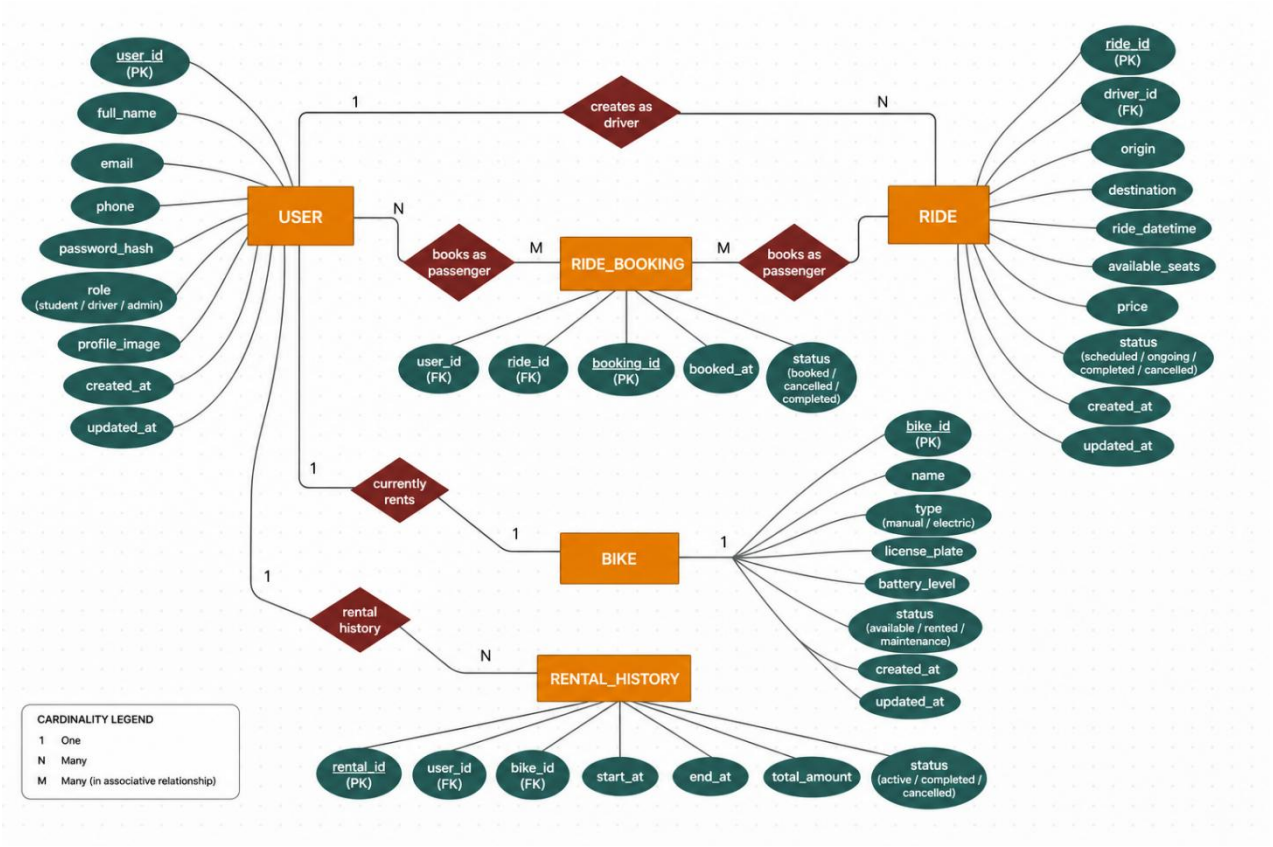
JSON Web Tokens (JWT)

Bcrypt.js





6. ER Diagram:



The Eco-Share platform utilizes a relational data structure mapped via MongoDB and Mongoose. The database consists of three primary entities: User, Ride, and Bike.

User Entity: Stores authentication details, role-based access levels (student, driver, or admin), and community metrics such as average ratings and total completed rides.

Ride Entity: Created by a User (acting as a driver) and contains journey specifics such as source, destination, fare, and seat availability. It holds a one-to-many relationship with the driver and a many-to-many relationship with the passengers (Users). It also securely logs timestamps and post-ride reviews.

Bike Entity: Represents the physical inventory of bicycles. It tracks the bike's location, type, current availability, and the user who is currently renting it. Additionally, it embeds a Rental History sub-document that maintains a log of past bookings, durations, and the specific users associated with those rentals.

This interconnected schema ensures data integrity while allowing for complex queries, such as fetching all active rides a student is participating in or calculating the rental history of a specific bike.

7. Database design

To manage the data in the application, Eco-Share is implemented using MongoDB, a NoSQL database. Using this database allows easy storage and retrieval of the application's data in the form of a JSON like structure. MongoDB is useful to support the fast fetching and handling of the data that is required by the features that facilitate real-time ride booking and coordination.

There are three main collections in the database; Users, Rides, and Bikes. The three collections are linked together using references, to maintain relationships between the entities.

7.1 User Collection

The user collection is used to store the information about each of the registered users of the system and handles the security and authentication of the user as well as the role of the user.

Fields:

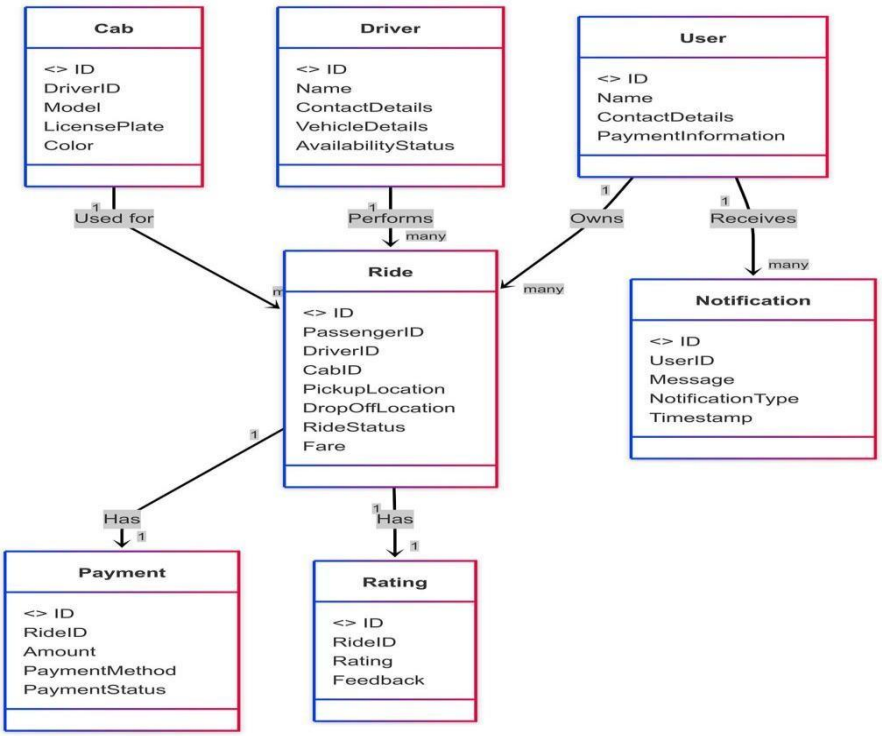
name: This field contains the full name of the user.

Email: This field contains the user ID of the user. This field should be unique for all users. Since the users of the college can only register using a college mail, this field can be a college email ID to ensure verified users.

Password: Stores the hashed value of the password. This password will be secured using the Bcrypt algorithm.

Role: This field defines the type of user that the user is. The roles will be classified into three types driver, passenger or administrato.

7.2 Rides Collection



This collection manages all the ride sharing related activities within the platform.

Fields:

driver: User object reference in the form of object ID pointing to the user document that created this ride.

Source: the starting point of the ride.

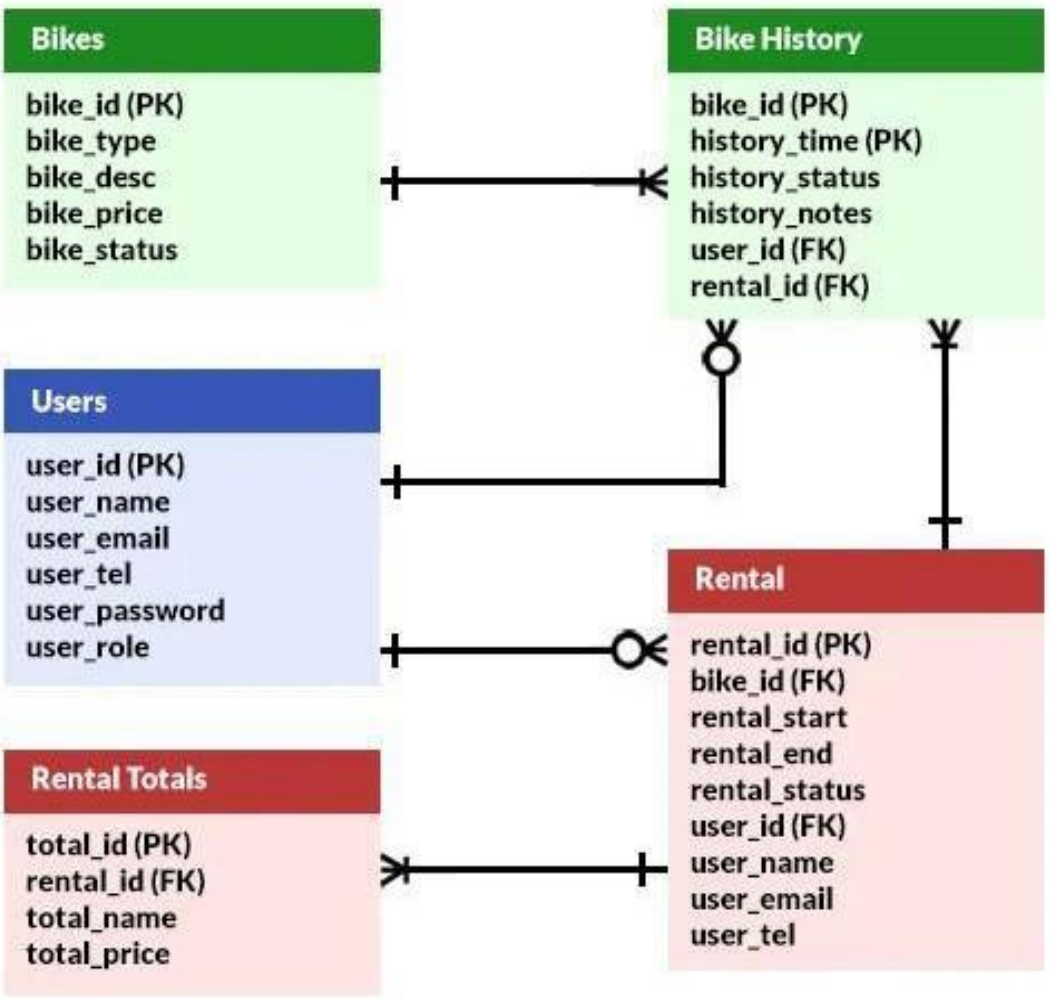
Destination: the ending point of the ride.

TotalSeats: This field stores the number of available seats in the car.

AvailableSeats: This field stores the current available number of seats for a ride.

Passengers: an array of User object references to the users who have boarded the ride.

7.3 Bikes Collection



This collection manages the booking and ownership of bikes used for intra-campus short travel.

Fields:

bikeName: this field holds the name or ID of the bike.

BikeType: the type or category of the bike like standard or electric, etc.

Location: current physical location of the bike on the campus.

IsAvailable: a boolean field which is true when the bike is not in use and can be booked.

CurrentUser: This is a User object reference pointing to the user currently occupying and hence possessing the bike.

Summary of the relationship

A User can have many Rides. (driver)

A User can take many Rides. (passengers)

A User can book many Bikes.

A Ride can be driven by a single user.

8. Implementation Highlights

8.1 Backend: Ride Booking Logic

The ride booking feature has been developed on the backend using Node.js and Express.js with MongoDB as the database. The controller for users joining the ride is provided below:

// ridesController.js

```
export const joinRide = async (req, res) => {
  const { id } = req.params;

  const ride = await Ride.findById(id);

  if (ride.availableSeats > 0) {
    ride.passengers.push(req.user._id);
    ride.availableSeats--;

    await ride.save();

    res.status(200).json({
      message: "Successfully joined ride!"
    });
  } else {
    res.status(400).json({
      message: "No seats available."
    });
  }
};
```

How does it work?

1. Retrieves ride using the ride ID from the params

2. Checks if seats are available (greater than 0)

3. If yes:

- Adds the logged in user ID to the passengers list

- Decreases the number of available seats by 1

- Updates the document in the database

- Returns a successful response to the frontend

4. If no:

- Returns a failure message to the frontend

Viva Point / Advanced Consideration:

In a real-world application, atomicity would need to be maintained, for example using findOneAndUpdate. Multiple users booking seats at the same time could otherwise race against each other if using separate findById and then save commands.

8.2 Frontend: Responsive UI

The frontend has been built using React along with Tailwind CSS.

Tailwind CSS's features that were used on the frontend:

- Mobile first approach: The UI dynamically adjusts its layout according to the size of the device being used to provide a seamless experience to the user (specifically the students).

- Utility based styling: No need to write any custom CSS as all classes were built with Tailwind's built in classes which are based on utility classes which is why development is very fast and design is consistent.

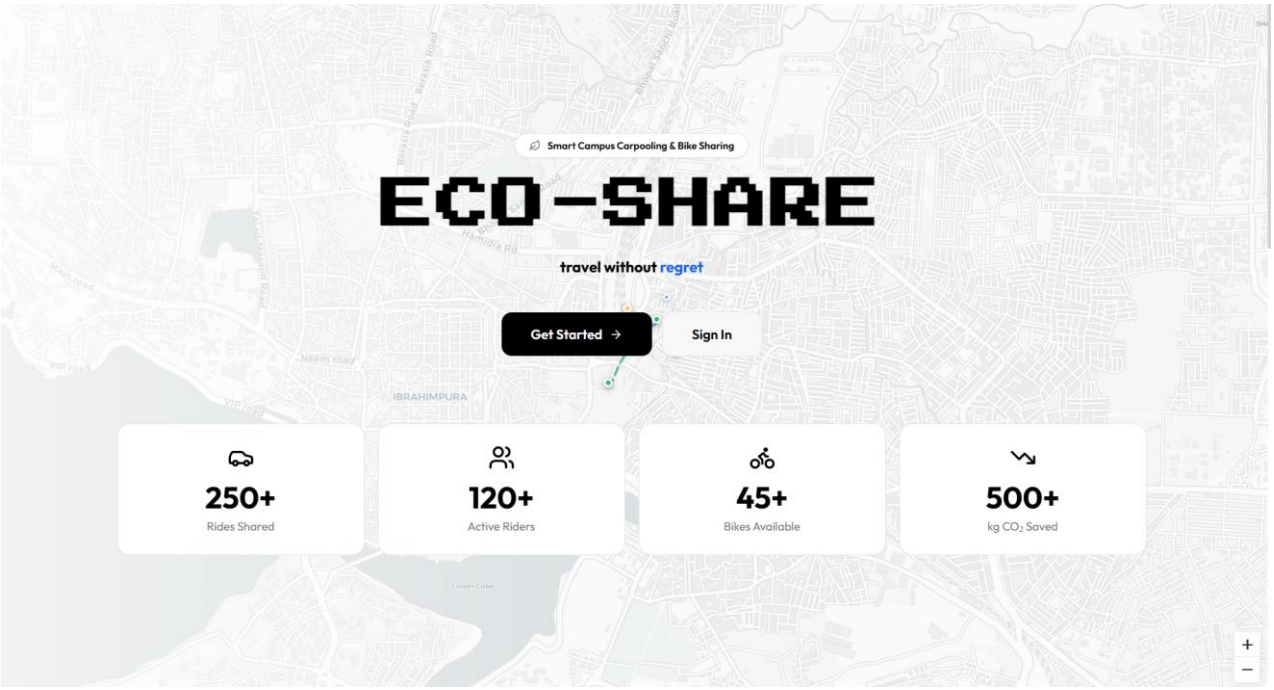
- Responsive components: Components like the Live Trips board, user dashboards, and booking interface adapt smoothly to different screen sizes.

- Better User Experience: Easy to navigate, clean UI design and quick rendering speed through React + Vite.

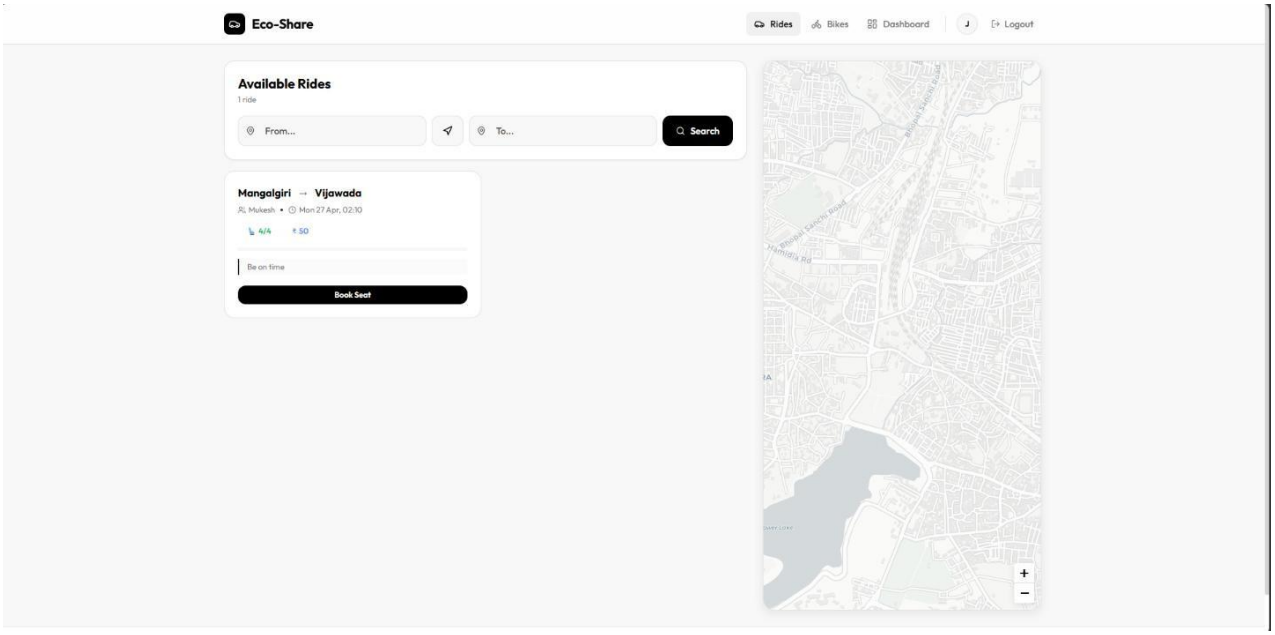
9. Results & Snapshots

Here is a display of the visual outcomes of the Eco-Share application, explaining what each function performs and illustrating how a user would access each module. Each screenshot focuses on ride-sharing and booking of bicycles.

9.1 Landing page



9.2 Dashboard – Active rides



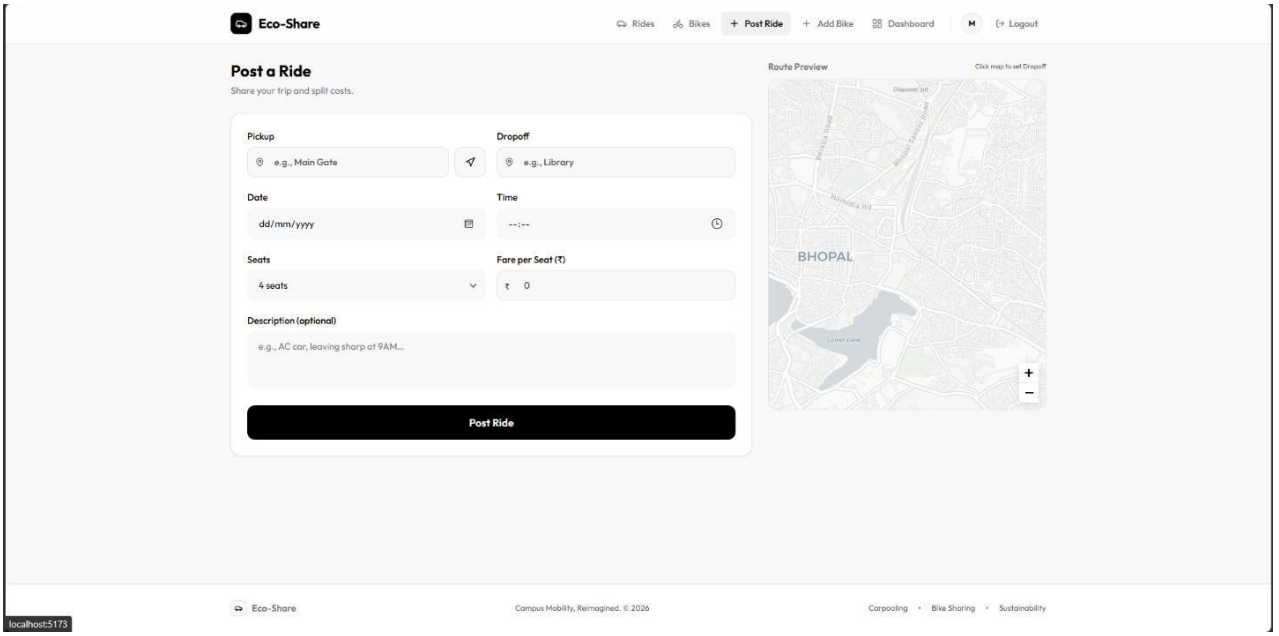
Here, all of the active rides, from those who want to share the ride from a particular destination to the destination where they are traveling to, are displayed. Users will have:

• The ability to see all of the ride information like source, destination and how many seats are available.

• The ability to see rides that would suit them easily.

• The real time availability of rides.

9.3 Create Ride Form

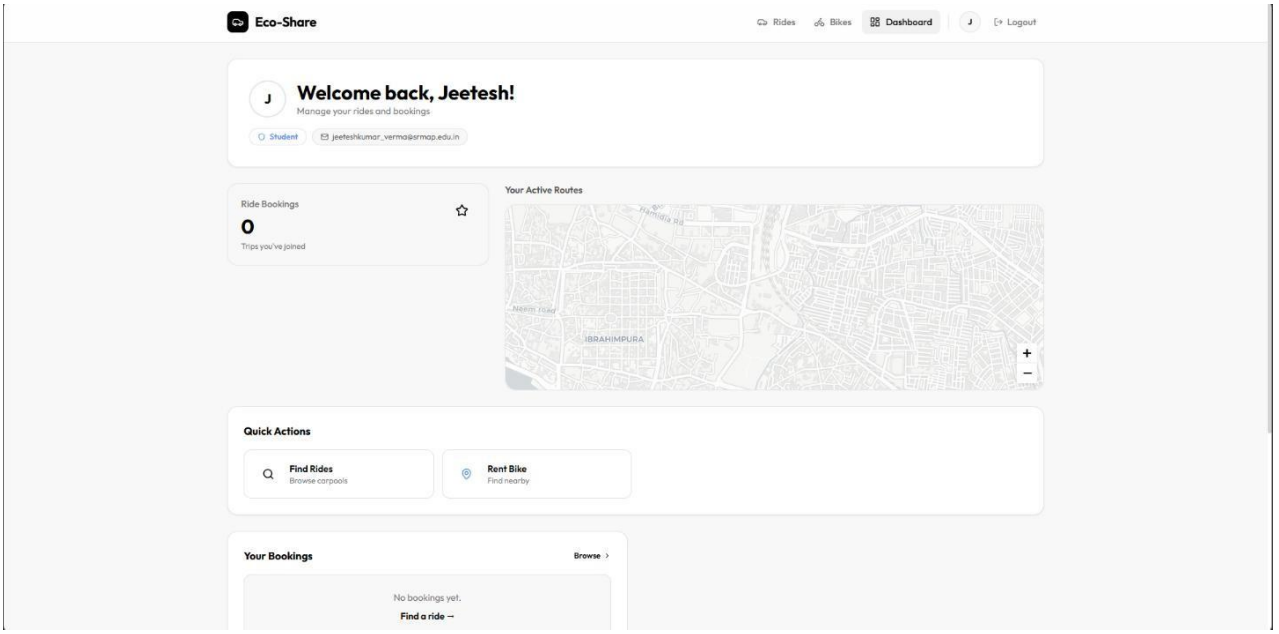


This page shows how a user (driver) can create new rides:

• The form captures source and destination along with details of available seats and time of journey.

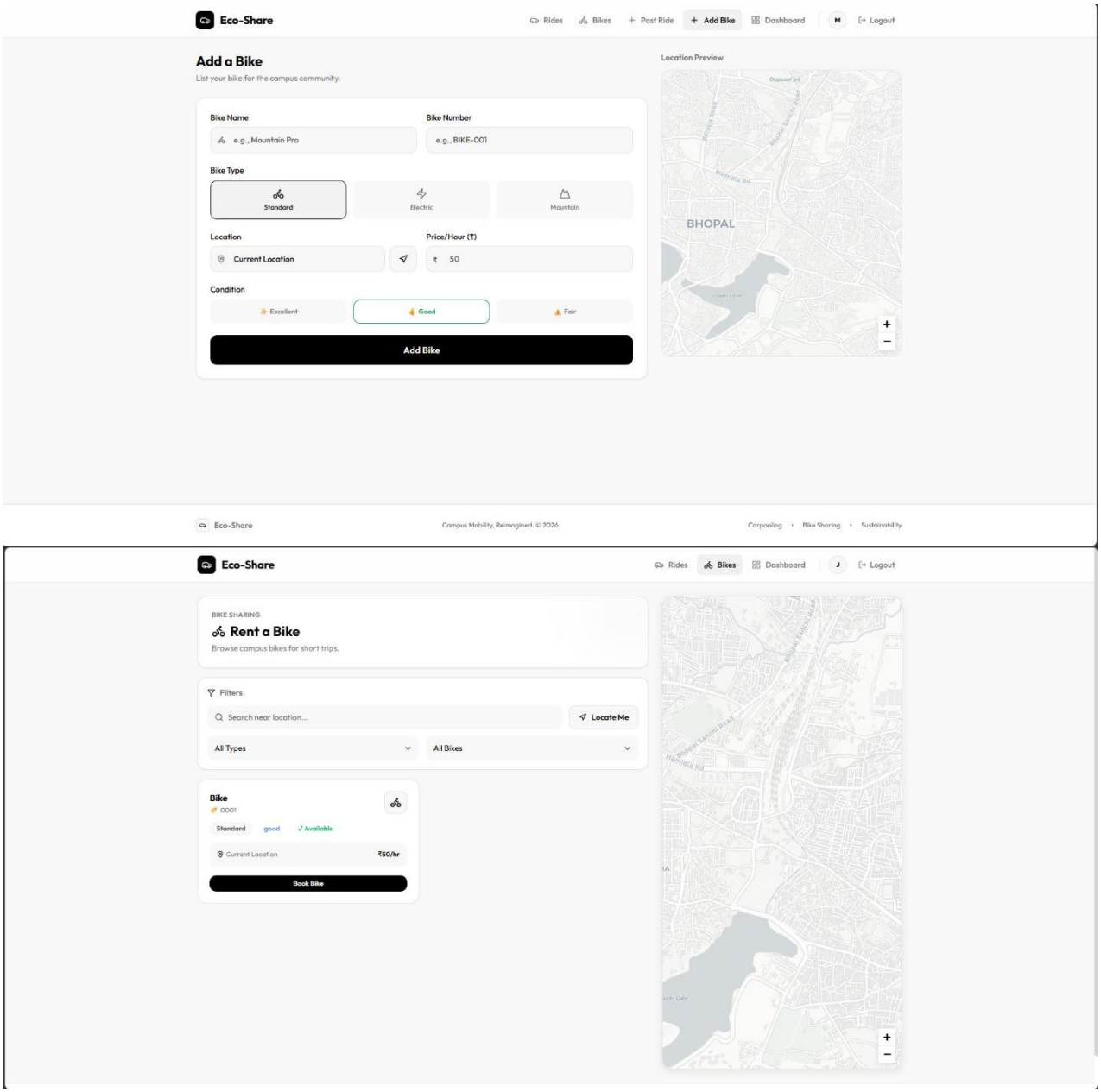
• The form has been made to be fast responsive, and user friendly for quick ride creation.

9.4 Student Dashboard



The Student Dashboard provides a user-friendly interface for passengers to explore and manage their commuting options. It allows students to view available rides in real time, join rides based on their preferred route, and book bikes for short-distance travel within the campus. The dashboard also displays booking history and active trips, enabling students to track their travel activities efficiently. Its intuitive design ensures quick navigation and seamless interaction with the platform.

9.5 Bike Listing Pag



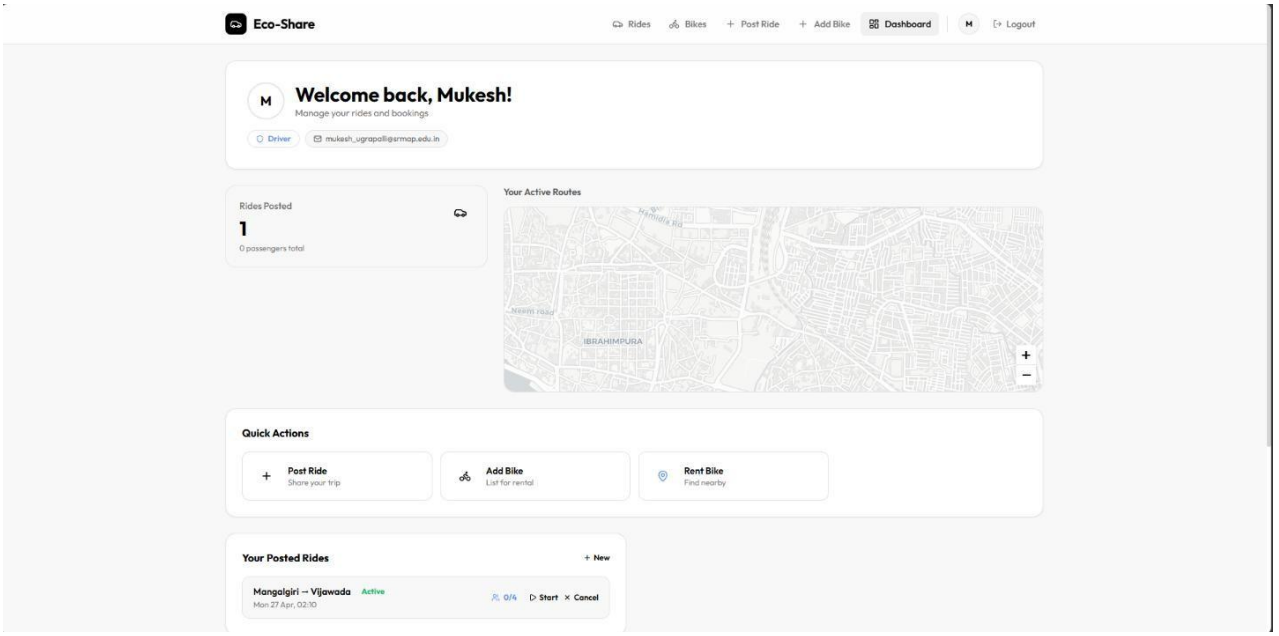
This displays all the available bicycles in campus, Users will have:

- The ability to see details about each bicycle, where is it placed, how many seats are available for travelling and how many seats have been taken.

• The ability to instantaneously book any bicycle.

• The ability to see if the bicycle has been taken or not.

9.6 Driver Dashboard



The Driver Dashboard is designed for users who offer rides to others. It enables drivers to create and manage ride listings by specifying details such as source, destination, and available seats. Drivers can monitor the number of passengers who have joined their ride and track seat availability dynamically. The dashboard also provides access to past rides and current trip status, helping drivers efficiently manage their ride-sharing activities.

10. Conclusion & Future scope.

The Eco-Share has fulfilled the needs of the student's commute within the college premises by providing an interface, centralized place, a trusted and effective mode of transportation. Integrating carpooling with bicycle-sharing brings down the traveling costs, reduces the pollution caused by various individual vehicle journeys and contributes to an eco-friendly mode of transportation.

Use of MERN-stack provides the web application with the necessary responsiveness, scalability, real-time transactions between the users. Integrating features like posting a ride, real-time seat management, bike booking and personalized dashboards increase the usability and effectiveness. By confining access to college registered users, the system maintains safety and builds trust among the users.

The Eco-Share clearly proves that technology can be utilized efficiently to create an eco-friendly, smart and a community driven transportation mode for college users.

Future work:

The existing features are satisfactory, but with few amendments the performance and user satisfaction can be significantly increased. Some of the planned future enhancements are:

GPS integration: With this enhancement a user will be able to track his ride, estimated time of arrival and other similar features, the driver and passenger will be better connected through live location tracking.

Payment Gateway Integration: Using an online payment system, we can automate the ride fee division process among the carpool participants, thus making the transaction easier and also enabling quick transactions for bike rentals.

In-app chat feature: The system will allow direct chat facility to the drivers and passengers, thus eliminating the need to use other communication apps for the same.

Rating and Feedback: users can rate their respective rides/drivers and provide feedbacks. This feature helps the users to gain the confidence and also provides an avenue for improvisation for better service and efficiency.

Push Notification: This will allow users to be aware about ride requests, confirmed rides, cancelled rides etc in real time.

Eco-Share lays the foundation for a smart campus mobility ecosystem and has the potential to be expanded into larger communities, contributing to the broader vision of sustainable and shared transportation systems.

11. Sources

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Demo Video Link:

https://drive.google.com/drive/folders/12FI2FtWujsgsg405dhL3v1NbCXkkSzgE?usp=sharing

Code Drive Link:

https://drive.google.com/drive/folders/1KW6azJwUd_OCTEKpJTCnMLCgCfBJ_fEB?usp=sharing